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O4 Olympiad geometry: problems

Thirty problems in three levels. For every problem, first draw a large accurate diagram (compass and ruler, or GeoGebra). Write down a final answer for every computation and a complete argument for every "prove" problem. Hints are at the end; full solutions are in solutions.md.

Level 1 — Kangoeroe style (wizPROF)

1. In triangle ABC, $AB = AC$ and $\angle A = 40^\circ$. The bisector of angle B meets AC at D. Find $\angle BDC$. *Topic: angle chasing*

2. In triangle ABC, $AB = AC$. Point D lies on side AC such that $AD = BD = BC$. Find $\angle A$. *Topic: angle chasing, isosceles triangles*

3. $ABCD$ is a cyclic quadrilateral (vertices in this order on the circle) whose diagonals meet at P. Given $\angle ADB = 40^\circ$ and $\angle DBC = 25^\circ$, find $\angle APB$. *Topic: inscribed angles*

4. Five points A, B, C, D, E lie in this order on a circle (not necessarily equally spaced). Draw the five-pointed star $A \to C \to E \to B \to D \to A$. Find the sum of the five angles at the tips of the star, $\angle CAD + \angle DBE + \angle ECA + \angle ADB + \angle BEC$. *Topic: angle chasing, exterior angles*

5. A right triangle has legs 3 and 4. A square is placed inside it so that one corner of the square is at the right angle, two sides of the square lie along the legs, and the opposite corner lies on the hypotenuse. Find the side of the square. *Topic: similar triangles*

6. In triangle ABC, $AB = 6$, $AC = 9$ and $BC = 10$. The bisector of angle A meets BC at D. Find BD. *Topic: angle bisector theorem*

7. Two chords AB and CD of a circle intersect at P. Given $AP = 4$, $PB = 6$ and $CP = 3$, find the length of CD. *Topic: power of a point*

8. A circle has centre O and radius 5. Point P satisfies $PO = 13$. A line through P meets the circle at A and B, with A between P and B, and $PA = 9$. Find AB. *Topic: power of a point, tangents*

9. A triangle has sides 13, 14 and 15. Find its area, its inradius r and its circumradius R. *Topic: areas, Heron, rs, $abc/4R$*

Level 2 — NWO first/second round style

10. In triangle ABC, point D lies on BC with $BD : DC = 1 : 2$, and E is the midpoint of AD. Line BE meets AC at F. Find $AF : FC$ and $BE : EF$. *Topic: Menelaus, area ratios*

11. In triangle ABC, $AB = 5$, $AC = 6$ and $BC = 7$. Find the length of the median from A. *Topic: median length, Stewart*

12. In triangle ABC, $\angle A = 60^\circ$, $\angle B = 45^\circ$ and $BC = 6$. Find AC, AB and the area of the triangle in exact form. *Topic: sine rule with $2R$, areas*

13. Two circles meet at P and Q. A common tangent touches the first circle at T_1 and the second at T_2. Prove that the line PQ passes through the midpoint of T_1T_2. *Topic: power of a point, radical axis*

14. In an acute triangle ABC, BE and CF are altitudes (E on AC, F on AB). Prove that B, C, E, F lie on one circle, and that $\angle AEF = \angle ABC$. *Topic: cyclic quadrilaterals, Thales*

15. Triangle ABC has $AB = 13$, $BC = 14$ and $CA = 15$, and incentre I. Line AI meets the circumcircle again at M. Find AI and MI, and conclude that I is the midpoint of AM. *Topic: incentre/excentre lemma, trigonometry*

16. $ABCD$ is a parallelogram with area 60. M is the midpoint of CD, and BM meets the diagonal AC at P. Find the area of quadrilateral $APMD$. *Topic: similar triangles, area ratios*

17. The incircle of triangle ABC touches BC, CA, AB at D, E, F. Prove that AD, BE and CF pass through one point. *Topic: Ceva, tangent lengths*

18. A triangle has sides 5, 7 and 8. Find the angle opposite the side of length 7, the area, the inradius and the circumradius. *Topic: cosine rule, areas*

19. A square has side 2. A circle passes through two adjacent vertices of the square and is tangent to the opposite side. Find its radius. *Topic: power of a point or Pythagoras*

20. In triangle ABC, $\angle A = 70^\circ$, $\angle B = 60^\circ$ and $\angle C = 50^\circ$. Let H, O and I be the orthocentre, circumcentre and incentre. Find $\angle BHC$, $\angle BOC$ and $\angle BIC$. *Topic: triangle centres, angle chasing*

21. In triangle ABC, $AB = 4$, $AC = 6$ and $BC = 5$. Find the length of the bisector from A (from A to its intersection with BC). *Topic: angle bisector theorem, Stewart*

Level 3 — NWO final / BxMO style

22. In triangle ABC, $AB = AC$ and $\angle A = 20^\circ$. Point D lies on side AC with $\angle DBC = 60^\circ$, and point E lies on side AB with $\angle ECB = 50^\circ$. Find $\angle BDE$, with a synthetic proof. *Topic: angle chasing with an auxiliary point*

23. Let ABC be an acute triangle with orthocentre H, circumcentre O, and let M be the midpoint of BC.

(a) Prove that the reflection of H in line BC lies on the circumcircle.
(b) Prove that the reflection of H in the point M is the point of the circumcircle diametrically opposite A.
(c) Deduce that $AH = 2\,OM$. *Topic: orthocentre, cyclic quadrilaterals*

24. Let ABC be an acute triangle with altitudes AD, BE, CF and orthocentre H. Prove that H is the incentre of triangle DEF. *Topic: cyclic quadrilaterals, triangle centres*

25. (Simson line.) Point P lies on the circumcircle of triangle ABC. Let X, Y, Z be the feet of the perpendiculars from P to the lines BC, CA, AB. Prove that X, Y, Z are collinear. *Topic: cyclic quadrilaterals, directed angles*

26. (Brahmagupta.) $ABCD$ is a cyclic quadrilateral whose diagonals are perpendicular and meet at P. The line through P perpendicular to AB meets CD at M. Prove that M is the midpoint of CD. *Topic: angle chasing, inscribed angles*

27. Let ABC be a triangle with three different side lengths. The external bisectors of the angles at A, B, C meet the lines BC, CA, AB at X, Y, Z. Prove that X, Y, Z are collinear. *Topic: Menelaus, angle bisector theorem*

28. Triangle ABC has $AB = 13$, $BC = 14$ and $CA = 15$. Find the distance between its circumcentre O and its incentre I. *Topic: triangle centres, coordinates*

29. Let ABC be an acute triangle with orthocentre H. Points E and F lie on the sides CA and AB. Prove that H has the same power with respect to the circle with diameter BE and the circle with diameter CF. Conclude that if these two circles meet in two points, the line through those points passes through H. *Topic: power of a point, radical axis*

30. In triangle ABC, the points D, E, F lie on BC, CA, AB with $BD : DC = CE : EA = AF : FB = 1 : 2$. The segments AD, BE, CF bound a small triangle in the middle. What fraction of the area of ABC is it? Prove your answer. *Topic: Menelaus, area ratios*

Hints

1. Find $\angle ABD$ first; $\angle BDC$ is an exterior angle of triangle ABD.
2. Call $\angle A = x$ and express every angle in terms of x.
3. $\angle ACB = \angle ADB$ and $\angle DAC = \angle DBC$; then use an exterior angle of triangle BPC.
4. Let AC and AD meet BE at P and Q; look at triangles APQ, PCE and QBD.
5. The small triangle above the square is similar to the whole triangle.
6. $BD : DC = AB : AC$.
7. $AP \cdot PB = CP \cdot PD$.
8. Compute the tangent length from P first.
9. Heron with $s = 21$; then area $= rs$ and $R = \frac{abc}{4 \cdot \text{area}}$.
10. Menelaus in triangle ADC with the line BEF; then in triangle BFC with the line AED.
11. $m_a^2 = \frac{2b^2 + 2c^2 - a^2}{4}$.
12. $\angle C = 75^\circ$ and $2R = \frac{6}{\sin 60^\circ}$.
13. Let PQ meet T_1T_2 at N and compute the power of N with respect to both circles.
14. Two right angles on the segment BC.
15. $\cos A = \frac{33}{65}$; find $\sin \frac{A}{2}$; $AI = \frac{r}{\sin(A/2)}$ and $MB = 2R \sin \frac{A}{2}$.
16. Triangles PAB and PCM are similar with ratio 2.
17. Tangent lengths are $s - a$, $s - b$, $s - c$; use the converse of Ceva.
18. The cosine rule gives $\cos = \frac{1}{2}$.
19. Put the centre on the axis of symmetry of the square and use Pythagoras.
20. $\angle BHC = 180^\circ - A$, $\angle BOC = 2A$, $\angle BIC = 90^\circ + \frac{1}{2}A$.
21. First BD and DC from the angle bisector theorem, then Stewart.
22. Take G on AC with $\angle GBC = 20^\circ$. Show that BEG is equilateral and $GD = GB$.
23. (a) $\angle BHC = 180^\circ - A$. (b) $BHCH'$ is a parallelogram; show $\angle ABH' = 90^\circ$.
24. Find the cyclic quadrilaterals with diameters BH and CH; show $\angle HDF = \angle HDE = 90^\circ - A$.
25. $PXCY$ and $PZAY$ are cyclic (Thales). Compare $\measuredangle PYX$ and $\measuredangle PYZ$.
26. In right triangle APB with altitude PF: $\angle APF = \angle ABP$. Then show triangle MPC is isosceles.
27. The external bisector divides the opposite side externally in the ratio of the adjacent sides.
28. Coordinates $B = (0, 0)$, $C = (14, 0)$, $A = (5, 12)$; or Euler's formula $OI^2 = R^2 - 2Rr$.
29. The foot of the altitude from B lies on the circle with diameter BE.

30. Show that each cevian is cut in the ratio $3 : 3 : 1$; then the triangle with side AB and third vertex $AD \cap BE$ has area $\frac{2}{7}[ABC]$.